

Supplementary Material

Long-context DNA models do not use the allele: an audit of DNALongBench’s eQTL task

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This supplement reports the per-tissue, per-fold and per-seed numbers behind the main text, the construction of eQTLP-v2 and the clinical set, and the released-code defects with file and line references. Every table is generated from the result files by `analysis/make_appendix_tables.py`.

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A The published protocol, per tissue

Table S1 gives the per-tissue view of the published eQTLP split that §2 of the main text summarises.

Three things to read off it. First, **distance alone** scores 0.688–0.828 per tissue against published foundation-model numbers of 0.48–0.58: the zero-training baseline wins in every tissue. Second, the **test sets are tiny** — 11–34 positives — so the 95% CIs on the distance baseline are 0.10–0.19 wide, and no published FM number is separable from 0.5 at this sample size. Third, the **split is not stratified and not disjoint**: positive rates fall from 9–21% in train to 3–5% in test, every test gene already appears in training, and 82–92% of test pairs have a training variant within 100 kb.

We score distance from the released coordinates, $|\text{variant} - \text{TSS}|$, because that is what the padded input encodes: each input is right-padded with N to 450 kb, so the count of N tokens is a direct readout of the span. The released `distance_to_tss` column disagrees with the coordinates by more than 1 kb for 16% of negatives and no positives; scoring by that column instead gives a mean of 0.663, still above every FM.

B eQTLP-v2 construction and the full baseline grid

Construction. We pool the nine tissues into 25,600 unique variant–gene pairs and drop 111 with conflicting labels across tissues. Each positive is matched to one negative on log distance to the TSS and on the same side of the TSS, with a caliper of 0.05 on log distance; this yields 2,905 matched pairs (5,810 rows). Folds hold out whole chromosomes, balanced on positive count, giving 570–665 test positives per fold. We drop the published label-derived mask, which marks the positions of the positive eQTLs and is therefore a label leak.

On matching away distance. Distance to the TSS is genuinely predictive of regulatory effect, so matching removes real signal along with the shortcut, and the matched task is harder than the underlying biology requires. We match anyway for two reasons. The task’s stated aim is variant-effect prediction from sequence, and a benchmark that a distance baseline wins is not measuring that. And the published input hands the model the distance directly through its padding, so the shortcut needs no sequence understanding at all.

Table S1: Published eQTL test split, per tissue. **dist** is our zero-training distance baseline scored from the released coordinates, with a 95% bootstrap CI; **expert**, **hyena** and **cad-ph** are the published numbers. Every test gene appears in training.

Tissue	pos	neg	p train	p test	dist	95% CI	expert	hyena	cad-ph	gene	locus
Adipose_Subcutaneous	16	316	0.21	0.05	0.715	[0.55, 0.87]	0.736	0.513	0.541	100%	83%
Artery_Tibial	27	542	0.11	0.05	0.744	[0.64, 0.84]	0.741	0.479	0.547	100%	87%
Cells_Cultured_fibroblasts	11	350	0.17	0.03	0.688	[0.54, 0.82]	0.639	0.584	0.597	100%	82%
Muscle_Skeletal	16	397	0.11	0.04	0.798	[0.68, 0.90]	0.621	0.487	0.538	100%	87%
Nerve_Tibial	34	609	0.13	0.05	0.818	[0.76, 0.88]	0.683	0.511	0.588	100%	88%
Skin_Not_Sun_Exposed_Suprapubic	17	456	0.09	0.04	0.828	[0.73, 0.91]	0.710	0.471	0.586	100%	87%
Skin_Sun_Exposed_Lower_leg	26	607	0.10	0.04	0.694	[0.60, 0.78]	0.700	0.544	0.574	100%	91%
Thyroid	26	524	0.14	0.05	0.699	[0.58, 0.81]	0.612	0.529	0.527	100%	86%
Whole_Blood	17	305	0.18	0.05	0.776	[0.66, 0.88]	0.689	0.512	0.594	100%	84%

Table S2 reports the unmatched (**all**) grid alongside the matched one so the cost of matching is visible: on the random split with distance available, the composition baseline reaches 0.70, and the gap between that and the matched chromosome-holdout floor of 0.599 is what matching removes.

Composition of the matched set. 5,360 of 5,810 rows (92.3%) are single-nucleotide substitutions; the remaining 450 (7.7%) are short indels or MNVs, with a median length change of 2 bp and a maximum of 76 bp. Indels are balanced across labels (92.1% of negatives and 92.4% of positives are SNVs), so they cannot carry the label. Nor do they affect the embedding-shift result: the median relative shift restricted to substitutions is 1.52×10^{-4} (HyenaDNA) and 0.674×10^{-4} (Caduceus) against 1.52×10^{-4} and 0.675×10^{-4} over all pairs. Indels shift the representation slightly more (1.63 and 0.691×10^{-4}), as expected from changing more bases, but they are too few and too similar to move the pooled statistic.

C Per-fold ablation results

Table S3 is the per-fold data behind Figure 1a, and Table S4 sweeps the probe’s regularisation strength.

The **ref_copy** ablation refits nothing: it takes the classifier already fitted on $[e_{\text{ref}}, e_{\text{alt}}]$ and scores $[e_{\text{ref}}, e_{\text{ref}}]$, so any difference is attributable to the alternate allele alone. Across $C \in \{0.001, \dots, 10\}$, both reading orientations, linear and MLP heads, and 13 fold partitions, the gap never leaves the fourth decimal. The paired bootstrap 95% CI on the difference is $[-0.0002, 0.0002]$.

Orientation. HyenaDNA is causal and the published orientation puts the variant roughly 500 bp from the end, so in 71–86% of pairs only a few hundred positions follow it. Embedding the reverse complement instead, so that every downstream position can see the variant, changes AUROC by less than 0.01 and leaves the ablation gap unchanged. Bidirectional Caduceus behaves the same way, which rules out causality as the explanation.

Numerical precision. Re-embedding 400 pairs in full fp32 reproduces the bf16 shifts (difference-vector cosine 0.89–0.95 between the two precisions), and **diff_only** stays at chance. The null is not a rounding artifact.

D The CNN collapse

The benchmark’s eQTL head ends in a ReLU applied to the logits. Both logits can therefore clamp to zero, at which point the prediction is constant and every weight receives zero gradient: the run is dead and cannot recover. On the *published* split, 2 of 10 initialisations are dead before the first optimisation step, and after one epoch 5 of 10 (Whole_Blood) and 6 of 10 (Nerve_Tibial) seeds emit a constant score, i.e. AUROC exactly 0.5. The training script sets no random seed and evaluates the final-epoch model rather than the best checkpoint, so which of these outcomes is reported is a matter of chance. The CNN is also fed only the reference sequence, so it cannot use the allele even in principle.

Table S5 reports the repaired CNN — ReLU removed from the logits — on eQTL-v2, across 5 folds and 3 seeds. It sits near chance (0.517 mean) and below the composition floor, and the allele ablation moves it by -0.0002 on average, with a worst case of 0.0017 across the 15 runs.

Table S2: eQTLTP-v2 baseline grid. **all** is every pooled pair; **matched** is the distance-matched subset the paper uses. **random** allows locus memorisation; **gene** and **chrom** do not.

Version	Split	Model	AUROC	fold mean	fold SD	n	n_+
all	random	distance	0.790	0.790	0.008	25600	3254
all	random	gene_prior	0.703	0.703	0.002	25600	3254
all	random	composition	0.744	0.744	0.006	25600	3254
all	random	dist+comp	0.833	0.833	0.008	25600	3254
all	random	dist+comp+allele	0.834	0.835	0.006	25600	3254
all	random	dist+mask_count	0.810	0.811	0.007	25600	3254
all	gene	distance	0.790	0.768	0.034	25600	3254
all	gene	gene_prior	0.373	0.500	0.000	25600	3254
all	gene	composition	0.640	0.669	0.011	25600	3254
all	gene	dist+comp	0.788	0.779	0.031	25600	3254
all	gene	dist+comp+allele	0.790	0.780	0.029	25600	3254
all	gene	dist+mask_count	0.781	0.767	0.035	25600	3254
all	chrom	distance	0.790	0.770	0.055	25600	3254
all	chrom	gene_prior	0.387	0.500	0.000	25600	3254
all	chrom	composition	0.639	0.668	0.028	25600	3254
all	chrom	dist+comp	0.790	0.784	0.049	25600	3254
all	chrom	dist+comp+allele	0.790	0.783	0.047	25600	3254
all	chrom	dist+mask_count	0.780	0.768	0.047	25600	3254
matched	random	distance	0.500	0.500	0.015	5810	2905
matched	random	gene_prior	0.706	0.707	0.009	5810	2905
matched	random	composition	0.665	0.665	0.013	5810	2905
matched	random	dist+comp	0.670	0.670	0.017	5810	2905
matched	random	dist+comp+allele	0.670	0.670	0.018	5810	2905
matched	random	dist+mask_count	0.529	0.529	0.014	5810	2905
matched	gene	distance	0.500	0.488	0.039	5810	2905
matched	gene	gene_prior	0.432	0.500	0.000	5810	2905
matched	gene	composition	0.602	0.608	0.009	5810	2905
matched	gene	dist+comp	0.601	0.610	0.005	5810	2905
matched	gene	dist+comp+allele	0.596	0.604	0.006	5810	2905
matched	gene	dist+mask_count	0.450	0.463	0.014	5810	2905
matched	chrom	distance	0.500	0.495	0.054	5810	2905
matched	chrom	gene_prior	0.439	0.500	0.000	5810	2905
matched	chrom	composition	0.588	0.593	0.018	5810	2905
matched	chrom	dist+comp	0.582	0.590	0.029	5810	2905
matched	chrom	dist+comp+allele	0.590	0.599	0.031	5810	2905
matched	chrom	dist+mask_count	0.436	0.444	0.012	5810	2905

E Released-code defects

Found by reading the released repository and running its own loaders on real records. We did not re-run the published experiments; with these defects the released code cannot produce the published FM numbers at all, so those numbers must come from a version other than the one released. We report this as a reproducibility problem with the artifact.

1. **The eQTL loaders return empty arrays.** Both the HyenaDNA and the Caduceus eQTL datasets pad each input to 450,000 positions and then slice it with `[2028500:-2028500]`. Since $2,028,500 \times 2 > 450,000$, the slice is empty for every record, so the model receives a zero-length sequence.
2. **The token ids are wrong.** With the slice removed, the loaders take `argmax` over a one-hot encoding, which yields ids 0–3. In both pretrained vocabularies 0–3 are special tokens; the nucleotides A, C, G, T, N are ids 7–11. The model is fed padding and separator symbols where bases belong.
3. **The configs do not match the checkpoints.** The shipped configs specify 2 layers where the released checkpoints have 8 (HyenaDNA-medium-450k) and 16 (Caduceus-Ph-131k), and the pretrained-weights

Table S3: Per-fold test AUROC behind Figure 1a, and the `ref_copy` ablation on the identical fitted model. The CNN contributes 3 seeds per fold.

Model	Fold	Seed	<code>refalt</code>	<code>ref_copy</code>
CNN	0	0	0.4687	0.4688
CNN	0	1	0.5750	0.5752
CNN	0	2	0.4915	0.4916
CNN	1	0	0.5437	0.5437
CNN	1	1	0.4749	0.4756
CNN	1	2	0.4861	0.4855
CNN	2	0	0.5360	0.5359
CNN	2	1	0.5690	0.5694
CNN	2	2	0.5025	0.5026
CNN	3	0	0.5106	0.5113
CNN	3	1	0.5709	0.5713
CNN	3	2	0.5255	0.5273
CNN	4	0	0.4984	0.4985
CNN	4	1	0.4497	0.4492
CNN	4	2	0.5596	0.5598
Caduceus	0	0	0.5492	0.5491
Caduceus	1	0	0.5874	0.5874
Caduceus	2	0	0.5350	0.5349
Caduceus	3	0	0.5772	0.5772
Caduceus	4	0	0.5686	0.5690
Floor (composition)	0	0	0.5870	–
Floor (composition)	1	0	0.6144	–
Floor (composition)	2	0	0.5644	–
Floor (composition)	3	0	0.6096	–
Floor (composition)	4	0	0.5905	–
HyenaDNA	0	0	0.5908	0.5908
HyenaDNA	1	0	0.5225	0.5225
HyenaDNA	2	0	0.5719	0.5717
HyenaDNA	3	0	0.5694	0.5696
HyenaDNA	4	0	0.5250	0.5253

path is left unset, so a run started from these configs trains a much smaller randomly initialised model.

4. The CNN’s logit ReLU collapses training, as in Appendix D.

Table S4: Probe-strength sweep on the fixed chromosome folds. The `refalt-ref_copy` gap stays at the fourth decimal at every regularisation strength, and every probe sits below the floor.

Model	C	<code>refalt</code>	<code>ref_copy</code>	<code>diff_only</code>	floor	Δ
HyenaDNA	0.001	0.5704	0.5704	0.4968	0.599	-0.00003
HyenaDNA	0.01	0.5667	0.5667	0.4958	0.599	+0.00002
HyenaDNA	0.1	0.5559	0.5560	0.4953	0.599	-0.00006
HyenaDNA	1	0.5472	0.5472	0.4955	0.599	-0.00009
HyenaDNA	10	0.5373	0.5377	0.4955	0.599	-0.00040
Caduceus	0.001	0.5775	0.5775	0.4972	0.599	-0.00005
Caduceus	0.01	0.5693	0.5693	0.5031	0.599	-0.00003
Caduceus	0.1	0.5635	0.5635	0.5068	0.599	-0.00002
Caduceus	1	0.5476	0.5477	0.5055	0.599	-0.00012
Caduceus	10	0.5341	0.5347	0.5053	0.599	-0.00059

Table S5: End-to-end CNN on eQTL-v2, 5 folds $\times$ 3 seeds. **alt** is the real pair, **ref_copy** replaces the alternate allele with the reference, and **rand** substitutes a base that never occurred.

Fold	Seed	alt	ref_copy	rand	Δ
0	0	0.4687	0.4688	0.4687	-0.00006
0	1	0.5750	0.5752	0.5753	-0.00024
0	2	0.4915	0.4916	0.4915	-0.00000
1	0	0.5437	0.5437	0.5438	-0.00003
1	1	0.4749	0.4756	0.4751	-0.00069
1	2	0.4861	0.4855	0.4862	+0.00055
2	0	0.5360	0.5359	0.5357	+0.00008
2	1	0.5690	0.5694	0.5693	-0.00040
2	2	0.5025	0.5026	0.5026	-0.00002
3	0	0.5106	0.5113	0.5113	-0.00073
3	1	0.5709	0.5713	0.5710	-0.00030
3	2	0.5255	0.5273	0.5262	-0.00175
4	0	0.4984	0.4985	0.4986	-0.00008
4	1	0.4497	0.4492	0.4499	+0.00059
4	2	0.5596	0.5598	0.5592	-0.00022

F Clinical set construction and results by stratum

Construction. We take ClinVar pathogenic and likely-pathogenic SNVs with a review status of at least one star, annotate them against GENCODE, and keep only variants that are non-coding, more than 50 bp from any annotated splice site, outside every CDS of every transcript, and outside small structural RNAs. 652 of the survivors map to a gene TSS within 450 kb. Each is matched to a benign SNV passing the identical filters, on distance to the TSS and, where possible, within the same gene and the same location class; 555 of 652 pairs match on both gene and class.

Why evaluation-only. After the splice, CDS and structural-RNA filters, ClinVar leaves 409 regulatory positives at two stars and 1,169 at one star — too few, and too concentrated in a handful of genes, to train on without simply memorising those genes. No model is fitted on clinical labels; every score is transferred from eQTL-v2.

Results. The embedding shift, the transferred eQTL-v2 probe and that probe’s allele-only component all score 0.489–0.507 over the full set and 0.49–0.52 on the 285 pairs whose pathogenic member has two stars or better. The only stratum above chance is 5’UTR/promoter at 0.53–0.57, and that is a distance effect: matching there left pathogenic variants closer to the TSS (median 170 bp vs 450 bp), and distance alone scores 0.61 on that stratum.

Gene concentration. The set is dominated by a few well-studied genes, so Table S6 repeats the evaluation after dropping the three most frequent genes, after dropping the ten most frequent, and after keeping a single pair per gene. Every score stays at chance, so the null is not an artifact of gene composition.

Interpretation. Pathogenicity and expression effect are different targets: a variant can shift expression without being pathogenic, or be pathogenic through a mechanism unrelated to expression. This result therefore corroborates the eQTL finding on an independent label set rather than showing that the eQTL conclusion generalises to clinical pathogenicity.

Table S6: Clinical set, sensitivity to gene concentration. Dropping the most frequent genes, or keeping one pair per gene, leaves every score at chance.

Quantity	all	drop NF1/HBB/ATM	drop top 10 genes	one pair per gene
pairs	652	564	466	301
baseline:-dist	0.51	0.51	0.504	0.5
HyenaDNA:shift_fwd	0.503	0.501	0.496	0.495
HyenaDNA:shift_rev	0.506	0.504	0.498	0.496
HyenaDNA:eqtl_probe	0.503	0.506	0.497	0.495
HyenaDNA:eqtl_allele	0.502	0.484	0.472	0.486
Caduceus:shift_fwd	0.496	0.501	0.499	0.497
Caduceus:shift_rev	0.5	0.502	0.497	0.5
Caduceus:eqtl_probe	0.507	0.506	0.497	0.5
Caduceus:eqtl_allele	0.489	0.483	0.482	0.488